EXECUTIVE BRIEFING

STRATEGIC TRANSFORMATION BLUEPRINT // ENERGY ENTERPRISE GIANTS

Agentic Transformation for India's Energy Giants

Operationalizing AI from the control room to the boardroom.

Gemini Enterprise | *"Calibrate wireline cable stretch on Well A-12 and execute SciPy depth alignment across Runs 1–3..."* | Execute Run ↵

FULL VALUE CHAIN
Upstream · Midstream · Downstream

SOVEREIGN GOVERNANCE & TRUST
MeitY Compliant · Identity-Governed Access · Immutable Audit

DETERMINISTIC PHYSICS ARCHITECTURE
Autonomous Reasoning · Scientific Solvers · Deterministic Output

THE CAPITAL REALITY // HIGH-CONSEQUENCE PHYSICAL ASSETS

In energy, there are no low-consequence mistakes.

Oil & gas is an unforgiving, high-capital business. Multi-crore losses rarely stem from visible catastrophes—they leak quietly every single day through routine operational friction across siloed disciplines.



₹200–500 Cr

Single Deepwater Well

Rig spread rate, casing program, wireline logging suite & marine fleet sunk before first commercial oil is confirmed.

Billions

Acreage & Farm-In Bidding

Multi-year concession commitments and PSC acreage valuations negotiated against fragmented, legacy well archives.

₹30L – 1.2 Cr

Single Idle Rig Day (NPT)

Offshore drilling spread idling while petrophysicists and drilling superintendents manually reconcile wireline logs.

₹25–100 Cr

One Lost Refinery Day

CDU/FCC unplanned shutdown or off-spec crude run gross profit margin loss per operating day.

JAMES REASON'S SWISS CHEESE MODEL // SYSTEMIC LATENT RISK

● Baseline: 3 Aligned Vulnerabilities | Fortified: Autonomous Agents

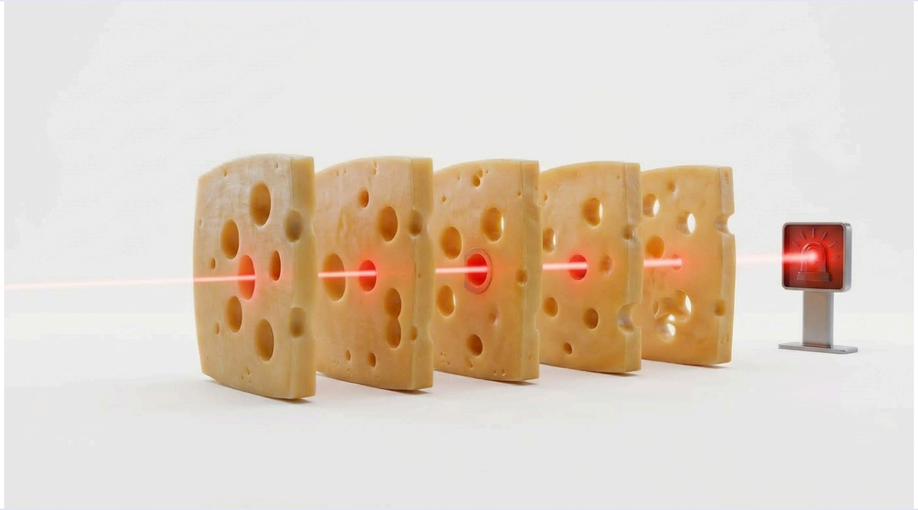
A major incident is never a single failure.

It's the quiet alignment of invisible gaps.

Disasters in high-consequence operations rarely stem from a single colossal error, but from the alignment of multiple vulnerabilities across siloed disciplines.

BARRIER TOPOLOGY // 4 DEFENSIVE LAYERS

LATENT HAZARD PATH: UNMONITORED CROSS-DISCIPLINE VULNERABILITIES



LAYER 01 // ARCHIVE — Vulnerability 1

Buried 2004 PDF Mudlog

Unindexed Scan · Overlooked Kick

LAYER 02 // PETROPHYSICS — Vulnerability 2

1.5m Cable Stretch

Mechanical Drift · Uncalibrated

LAYER 03 // RIG HANDOVER — Vulnerability 3

4-Hour Handover Lag

3:00 AM Shift Change · Warning Lost

LAYER 04 // DRILLING TARGET — CRITICAL HAZARD

₹45 Cr Rig NPT

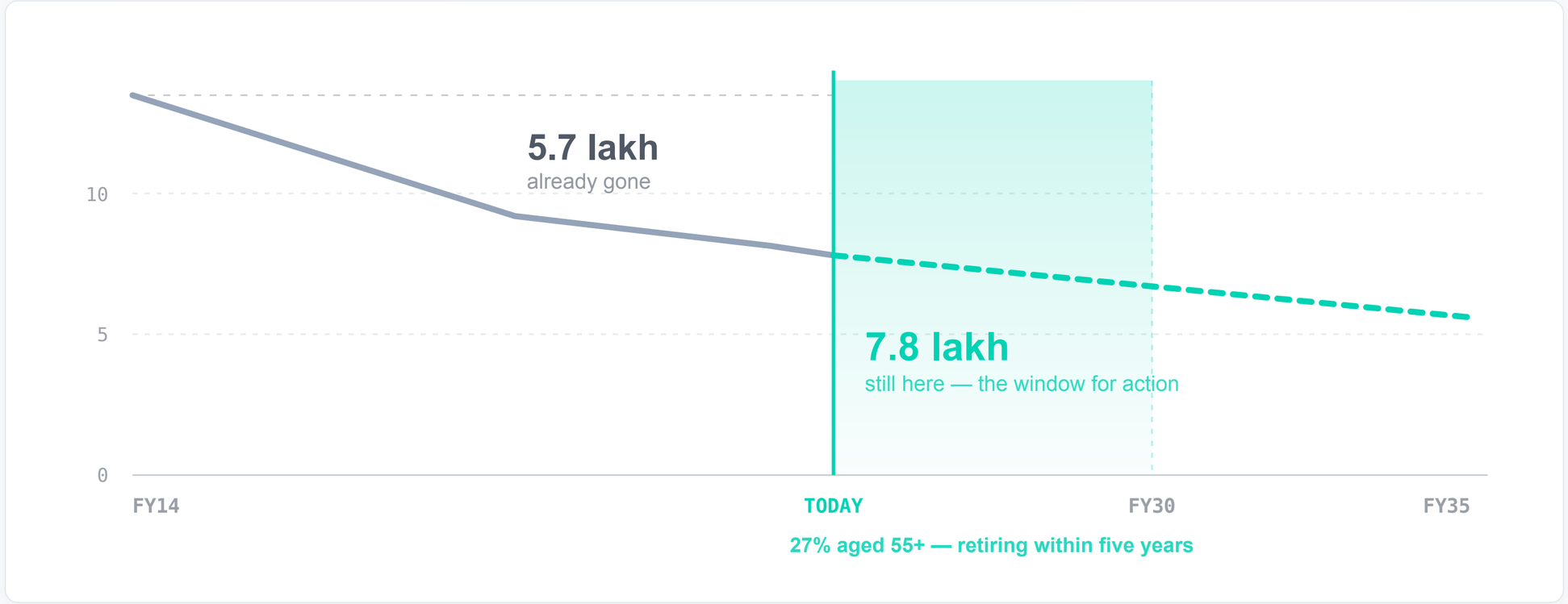
Casing in Gas Sand · Uncontrolled Kick

The Great Crew Change

As experienced staff leaves, latent vulnerabilities align more often—creating systemic operational risk across the organisation that will continue to compound unless we intervene.

INDIA · PUBLIC SECTOR REGULAR WORKFORCE (LAKH)

● 13.49 LAKH IN FY14 = 5.7 GONE + 7.8 STILL HERE



Public Enterprises Survey (DPE): 13.49 lakh FY14 · 9.2 FY20 · 8.14 FY24 · 7.8 FY25 · FY26–FY35 projected at trend

Source: pib.gov.in

Rising operational risk due to departing experienced personnel. Up to 50% of the global energy workforce is nearing retirement, while across Indian PSUs, 42% of regular employees have retired over the past decade alone—with 27% of the remainder exiting by FY30.

THE NEED // ARCHITECTURAL REQUISITES // THE DEFENSIVE PLUG

AI Agents provide targeted solutions to seal micro-vulnerabilities.

THE FIVE DEFINING ATTRIBUTES // NON-NEGOTIABLE FOR MISSION-CRITICAL OPERATIONS

01 // FOCUS 01

Surgical

Targets the exact seam where the operational vulnerability lives.

02 // TRIGGER 02

Autonomous

Triggers instantly on live operational data, guarding handovers 24/7 without human delay.

03 // CONTEXT 03

Intelligent

Understands domain context and cross-discipline data.

04 // CONTROL 04

Governed

Identity and access management, an immutable audit trail, and a human-in-the-loop veto.

05 // PHYSICS 05

Deterministic

Grounded in deterministic physics and exact mathematical solvers.

WHERE IT LIVES // THE COEXISTENCE PARADIGM

The future of the energy sector will be in the coexistence of domain monoliths and AI Agents.

PROVEN DOMAIN ENGINES

Domain Monoliths

Deep Domain Logic & Enterprise Truth

+

TARGETED WORKFLOW INTELLIGENCE

AI Agents

Targeting High-Risk Gaps & Optimization Value

=

THE UNIFIED OPERATING MODEL

Agentic Organization

Governed Fleet of Specialized Agents

FIELD VALIDATION // DETERMINISTIC WIRELINE SPLICING

The Petrophysical AI Agent

Eliminating hours of manual curve alignment with an autonomous subsurface physics agent.

01 THE CHALLENGE // MANUAL CURVE ALIGNMENT

Raw Well Logs Arrive Mismatched & Disconnected

Separate wireline logging passes never line up due to cable stretch and sensor drift. Today, senior petrophysicists spend **1–2 hours manually nudging and eyeballing curves at 3:00 AM**—where human fatigue risks multi-crore drilling depth errors.

02 WHAT IS NEEDED // TARGETED PHYSICS INTELLIGENCE

Automated Deterministic Alignment

A targeted solution that works alongside existing subsurface systems—normalizing tool calibration drift, running signal cross-correlation across the overlap, and locking in the exact depth offset where curve alignment is mathematically maximized.

03 WHAT THE AGENT BUILDS // UNIFIED COMPOSITE LOG

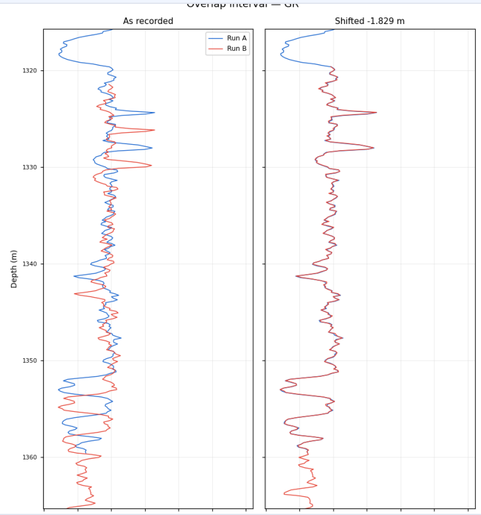
WITHIN SECONDS

Verified Composite Well Log Within Seconds

Within seconds, the AI Agent ingests raw passes, normalizes curve scales, and executes the optimal -1.8m shift where cross-correlation peaks (**r = 0.99**), delivering an audit-ready composite log for instant visual verification.

KANSAS WELL A-12 // AUTOMATED OVERLAP ALIGNMENT

PEAK CORRELATION: r = 0.99



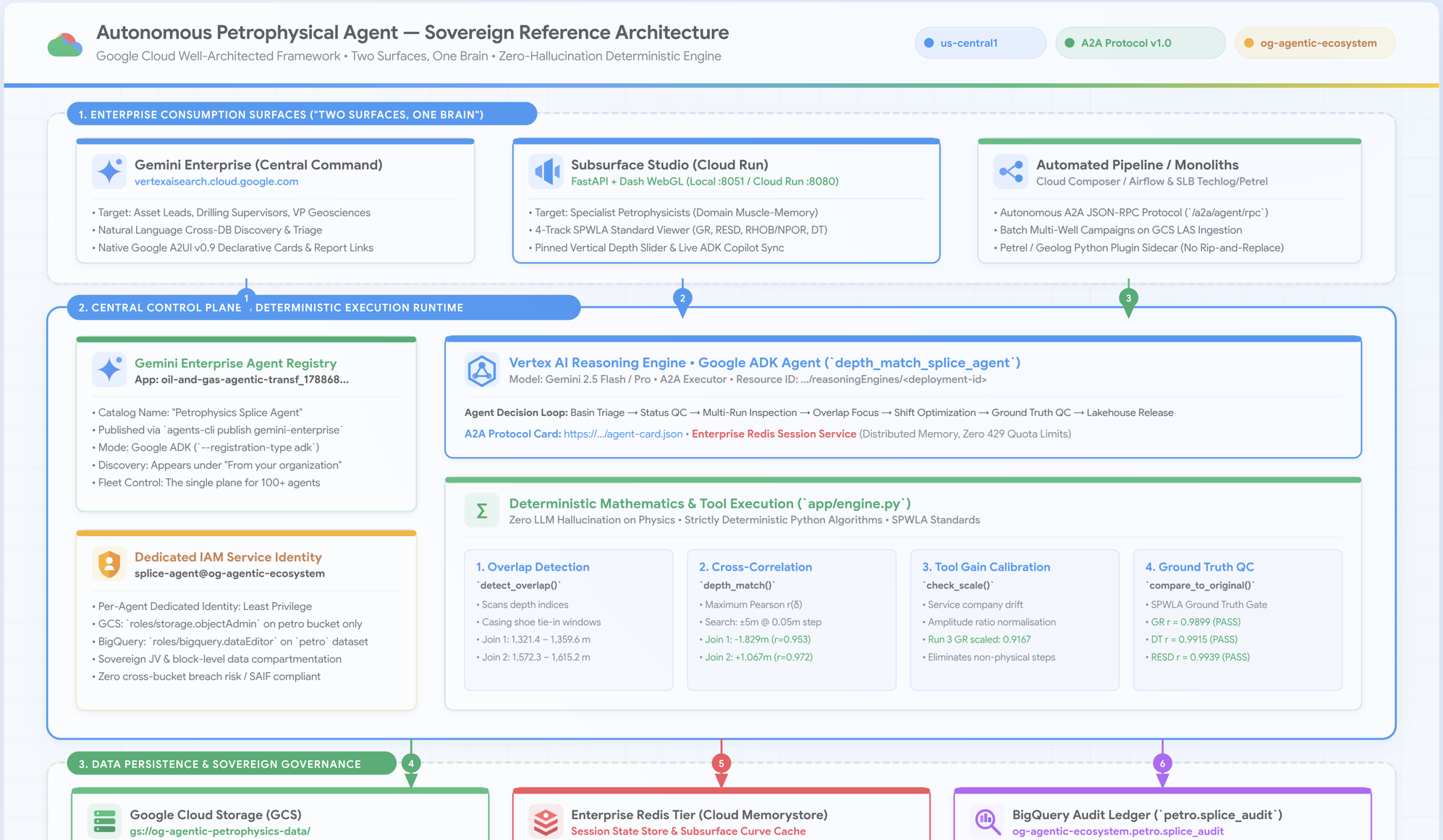
Raw Log: 1.83m offset (misaligned)

Agent Shifted -1.829m: Snapped into exact match within seconds

06 ARCHITECTURE // PRODUCTION AGENT & CENTRAL CONTROL PLANE

Architecture of the Petrophysical AI Agent

The production anatomy of our subsurface agent—built on **Google Agent Development Kit (ADK)**, powered by deterministic physics, and centrally governed as the reference blueprint for enterprise scale.



WORKFORCE & WORKFLOW MAP · 33 ROLES · 498 WORKFLOWS · 234 AGENTS

Agentic Transformation Across Workforce & Workflows

Mapping personas and workflows to identify and prioritise operational risks, friction, and vulnerabilities.

ANNUAL VALUE	PRODUCTIVITY	UPTIME	INTEGRITY	RECOVERY	AGENTS	AGENT PREPARES
₹3,654 Cr	₹1,497 Cr · 41%	₹845 Cr · 23%	₹300 Cr · 8%	₹1,011 Cr · 28%	234	335 of 498 workflows

33 personas · 498 workflows · ₹3,654 Cr/yr

All personas 33 | Subsurface 7 | Wells & Rig 5 | Field & Gathering 4 | Midstream 6 | Downstream Refining 5 | Cross-Cutting 6 | Build priority | Value map

WHO IMPLEMENTS IT — Agent prepares, human decides 335 · Existing system runs it 86 · Human, end to end 77 — BUILD QUEUE — Critical 36 · Friction 140 — *only work that is ours to build is marked*

WORKFORCE	VALUE / YR	15–16 WORKFLOWS PER ROLE — AGENT PREPARES · SYSTEM RUNS · HUMAN HOLDS														
Wellsite Supervisor P01 · Wells & Rig · 7 agents	₹135.4 Cr	P01W01	P01W02	P01W03	P01W04	P01W05	P01W06	P01W07	P01W12	P01W13	P01W14	P01W08	P01W11	P01W09	P01W10	P01W15
Well Integrity Engineer P02 · Wells & Rig · 8 agents	₹39.7 Cr	P02W01	P02W02	P02W03	P02W05	P02W06	P02W07	P02W08	P02W09	P02W14	P02W15	P02W04	P02W11	P02W12	P02W10	P02W13
Board Operator (Refinery) P03 · Downstream Refining · 7 agents	₹199.3 Cr	P03W01	P03W02	P03W03	P03W04	P03W05	P03W06	P03W07	P03W13	P03W14	P03W08	P03W10	P03W12	P03W09	P03W11	P03W15
Petrophysicist P04 · Subsurface · 8 agents	₹194.3 Cr	P04W01	P04W02	P04W03	P04W04	P04W05	P04W06	P04W07	P04W16	P04W08	P04W09	P04W10	P04W11	P04W14	P04W15	P04W12	P04W13
Petroleum Geologist P05 · Subsurface · 8 agents	₹148 Cr	P05W01	P05W02	P05W03	P05W04	P05W05	P05W06	P05W07	P05W13	P05W15	P05W16	P05W08	P05W09	P05W10	P05W11	P05W12	P05W14
Reservoir Engineer P06 · Subsurface · 7 agents	₹147.2 Cr	P06W01	P06W02	P06W03	P06W04	P06W05	P06W06	P06W07	P06W12	P06W14	P06W08	P06W09	P06W11	P06W13	P06W10	P06W15
Drilling Engineer P07 · Wells & Rig · 8 agents	₹119.5 Cr	P07W01	P07W02	P07W03	P07W04	P07W05	P07W06	P07W07	P07W12	P07W13	P07W14	P07W16	P07W08	P07W09	P07W10	P07W11	P07W15
Completions Engineer P08 · Wells & Rig · 7 agents	₹56.4 Cr	P08W01	P08W02	P08W03	P08W04	P08W05	P08W06	P08W07	P08W11	P08W14	P08W08	P08W09	P08W10	P08W13	P08W12	P08W15
Production Engineer P09 · Field & Gathering · 7 agents	₹209.2 Cr	P09W01	P09W02	P09W03	P09W05	P09W06	P09W07	P09W08	P09W09	P09W12	P09W04	P09W10	P09W11	P09W13	P09W14	P09W15
Lease Operator P10 · Field & Gathering · 7 agents	₹453.9 Cr	P10W01	P10W03	P10W05	P10W07	P10W08	P10W11	P10W14	P10W02	P10W04	P10W06	P10W09	P10W10	P10W12	P10W13	P10W15
Control Room Operator (Produ… P11 · Field & Gathering · 7 agents	₹124.7 Cr	P11W01	P11W02	P11W04	P11W06	P11W07	P11W09	P11W10	P11W13	P11W03	P11W11	P11W12	P11W05	P11W08	P11W14	P11W15

OPERATIONAL RESILIENCE // SYSTEMIC DEFENSE IN DEPTH

Structural Resilience of an Agentic Organisation

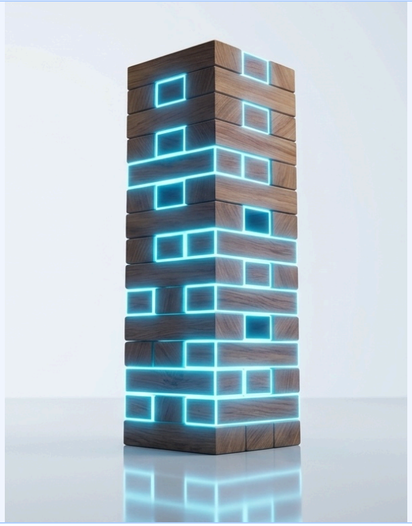
Operational vulnerabilities exist across both **workflows** and **workforce**. AI Agents eliminate these operational gaps to build a structurally resilient organization.



THE UNBRACED TOWER — HIGH SYSTEMIC FRAGILITY

Without AI Agents

Gaps across disconnected workflows and a stretched workforce leave mission-critical processes unmonitored and vulnerable to friction.



THE FORTIFIED MESH — STRUCTURAL RESILIENCE

With AI Agents

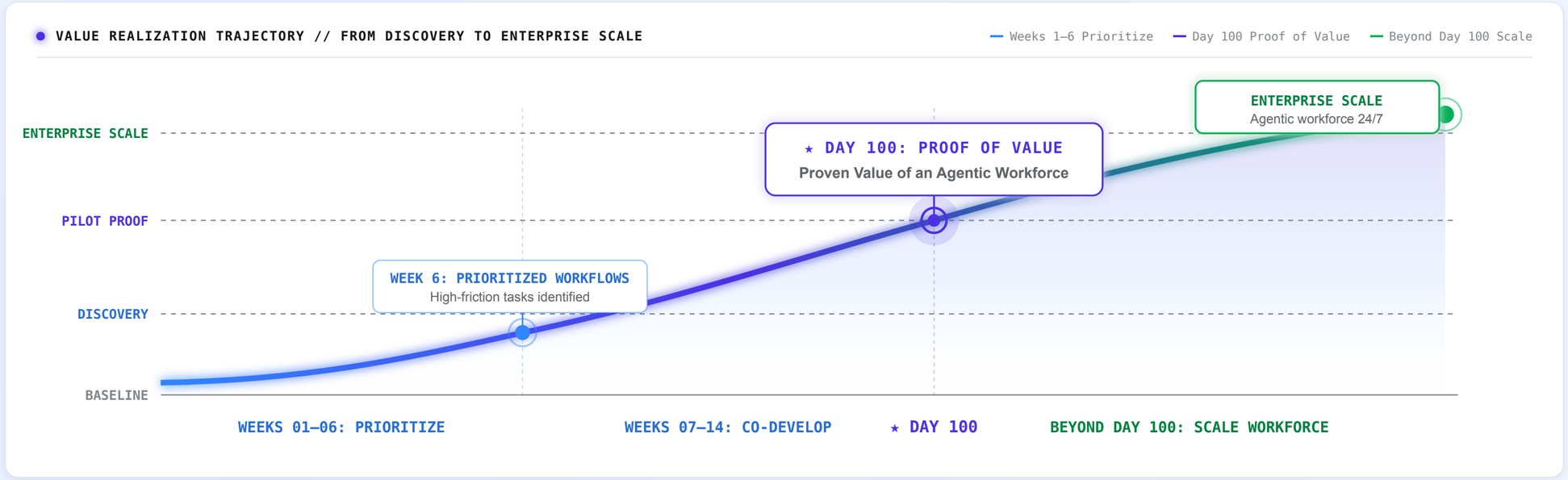
AI Agents augment both workforce and workflows—bridging operational gaps with 24/7 autonomous monitoring and exact solvers.

THE ROADMAP // 100-DAY PROOF OF VALUE • EXECUTIVE EXECUTION MODEL

100 Days to Proof of Value.

Building an agentic workforce to automate workflows.

Solving specific operational bottlenecks on your historical data—proving measurable value and mathematical accuracy in 100 days before scaling across the enterprise.



PHASE 01 // WEEKS 01–06

Prioritized Workflows

- Nominate initial domain leads and critical workflows
- Identify tasks with high operational friction, delay, or risk
- Integrate with existing workflows

OUTPUT: Prioritized Personas & Workflows

PHASE 02 // DAY 100 MILESTONE

Co-Develop & Prove Value

- Co-develop targeted agents alongside your domain specialists
- Validate mathematical accuracy against real historical data
- Quantify Agent ROI before enterprise rollout

OUTPUT: Proven Value of an Agentic Workforce

PHASE 03 // BEYOND DAY 100

Scale the Agentic Workforce

- Expand proven agents across relevant teams and departments
- Automate connected operational workflows 24/7
- Manage and govern agentic workforce

OUTPUT: Enterprise-Wide Automation